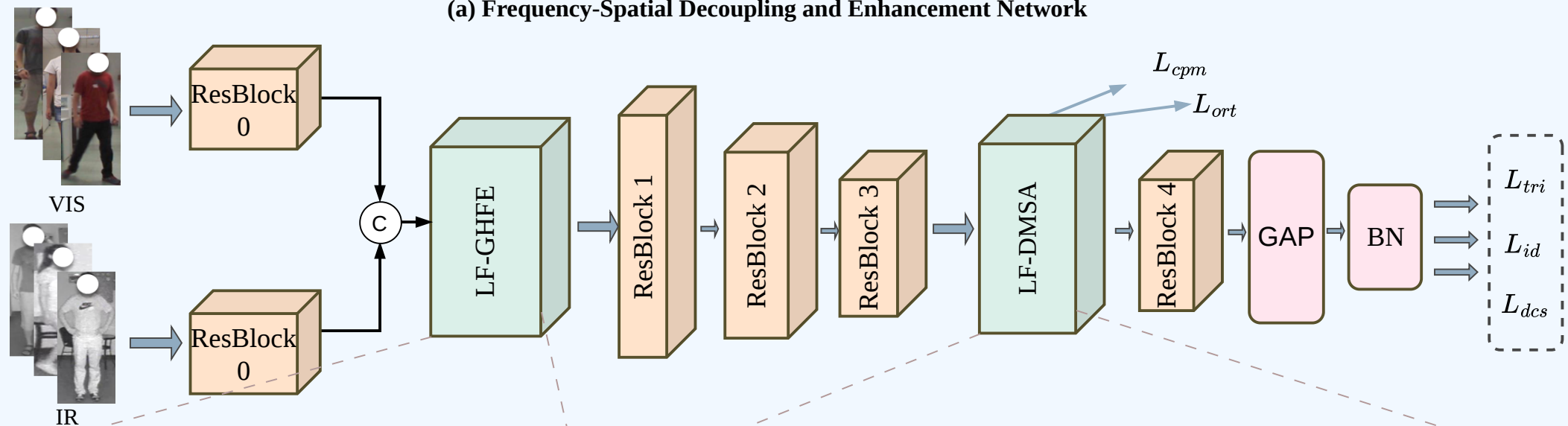
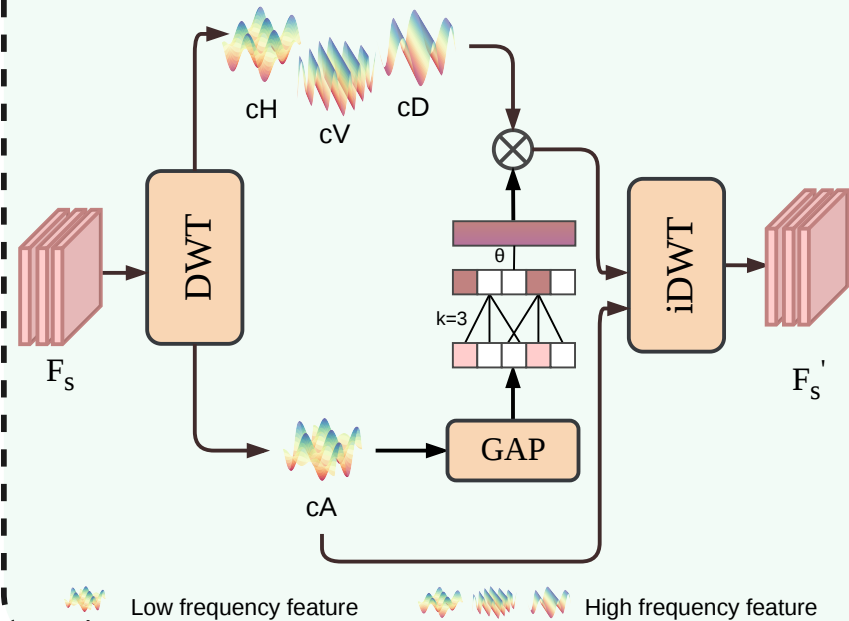


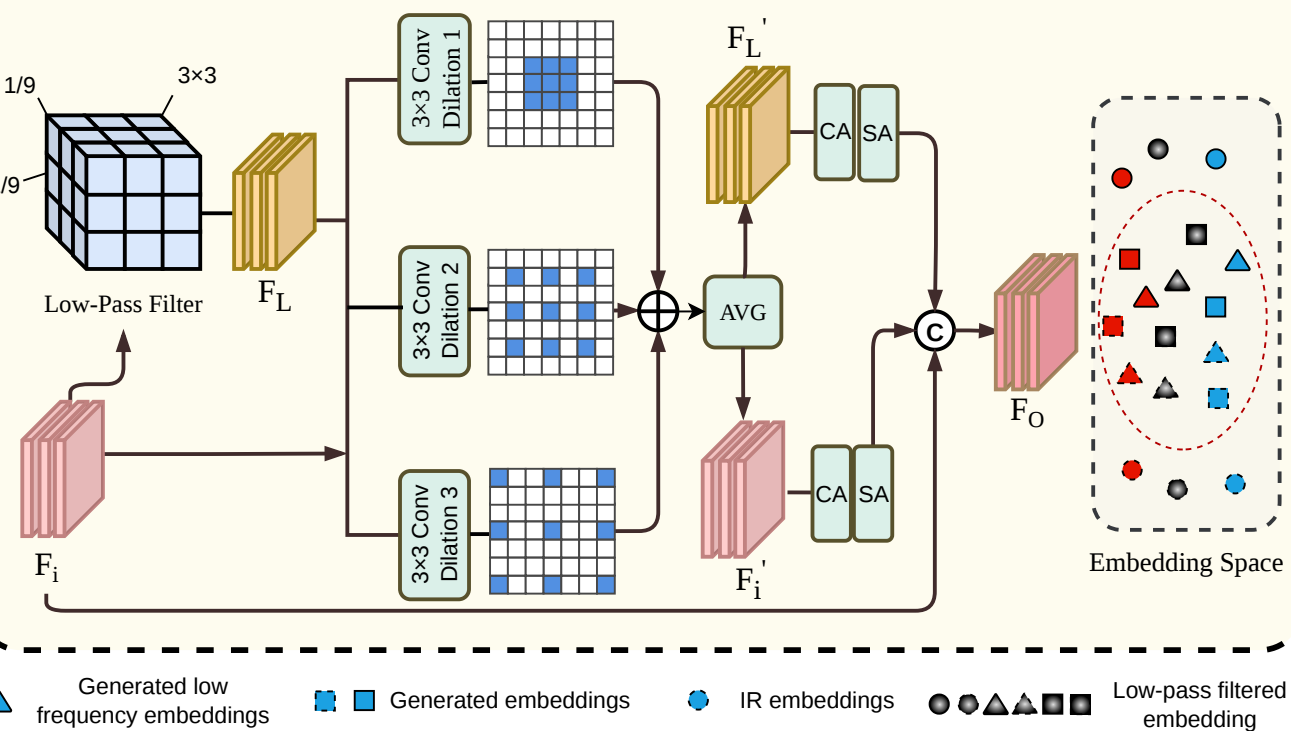
(a) Frequency-Spatial Decoupling and Enhancement Network



(b) Low-Frequency Guided High-Frequency Enhancement



(c) Low Frequency Decoupled Multiscale Spatial Alignment



$\otimes$ Multiplication $\oplus$ Addition $\textcircled{C}$ Concat ● VIS embeddings ▲ ▲ Generated low frequency embeddings ■ ■ Generated embeddings ● IR embeddings ● ▲ ■ Low-pass filtered embedding